

NCEA Level 2 Numeracy Knowledge Base

Overview

The NCEA Numeracy co-requisite ensures students can apply mathematics and statistics in everyday situations. From 2024, students must achieve 10 numeracy credits (plus 10 literacy credits) as a 20-credit co-requisite for any NCEA qualification.

Numeracy Standard

US32406: Use Mathematics and Statistics in a Range of Everyday Situations

Credits: 10

Level: 1

Assessment: External Common Assessment Activity (CAA)

Purpose:

Learners credited with this standard are expected to:

- Formulate mathematical and statistical approaches to solving problems in meaningful situations
- Use mathematics and statistics to meet numeracy demands of various contexts
- Explain the reasonableness of mathematical and statistical responses

Skills Required:

Number Operations:

- Fluently solve problems involving addition, subtraction, multiplication, division
- Understand relative size of numbers
- Interpret answers in context
- Work with fractions, decimals, and percentages
- Use ratios and proportions

Mathematical Relationships:

- Recognise and work with patterns
- Understand algebraic relationships
- Apply formulas to solve problems

Spatial Properties:

- Understand and use properties of shapes
- Work with representations of objects
- Apply geometric reasoning

Location and Navigation:

- Understand and use coordinate systems
- Read and interpret maps
- Apply navigation concepts

Measurement:

- Use appropriate units of measurement
- Measure with appropriate precision

- Convert between units
- Calculate perimeter, area, volume
- Work with time and rates

Statistics and Data:

- Interpret graphs, charts, and tables
- Calculate statistical measures (mean, median, mode)
- Understand data distribution
- Draw conclusions from data

Probability:

- Interpret situations involving chance
 - Calculate simple probabilities
 - Understand likelihood and risk
-

Assessment Criteria

- **Achieved:** Uses mathematics and statistics to meet the numeracy demands of a range of everyday situations
 - Graded as “Achieved” or “Not Achieved” only
 - No Merit or Excellence grades for unit standards
-

Key Numeracy Concepts

Number Sense

Operations:

- Addition and subtraction in context
- Multiplication for repeated groups
- Division for sharing and grouping
- Order of operations (BEDMAS)

Fractions, Decimals, Percentages:

- Converting between forms
- Finding percentages of amounts
- Percentage increase/decrease
- Comparing fractions

Ratios and Proportions:

- Simplifying ratios
- Scaling recipes or quantities
- Map scales
- Rate problems (speed, cost per unit)

Measurement

Length:

- mm, cm, m, km conversions

- Perimeter calculations

Area:

- Square units (cm^2 , m^2 , ha, km^2)
- Rectangle, triangle, circle areas
- Composite shapes

Volume and Capacity:

- mL, L conversions
- cm^3 , m^3
- Volume of rectangular prisms

Mass:

- g, kg, t conversions

Time:

- 12-hour and 24-hour time
- Duration calculations
- Time zones

Statistics**Data Types:**

- Categorical vs numerical
- Discrete vs continuous

Statistical Measures:

- **Mean:** $\text{Sum} \div \text{count}$ (average)
- **Median:** Middle value when ordered
- **Mode:** Most frequent value
- **Range:** Difference between highest and lowest

Graphs and Charts:

- Bar graphs
- Pie charts
- Line graphs
- Histograms
- Box plots

Interpretation:

- Reading data from graphs
- Identifying trends
- Making predictions
- Drawing conclusions

Probability**Basic Concepts:**

- Probability scale (0 to 1)
- Impossible, unlikely, likely, certain
- Experimental vs theoretical probability

Calculations:

- Probability = favourable outcomes ÷ total outcomes
- Complementary events ($P(\text{not } A) = 1 - P(A)$)
- Combined events

Geometry**Shape Properties:**

- Triangles, quadrilaterals, circles
- Angles (acute, right, obtuse, straight, reflex)
- Parallel and perpendicular lines

Transformations:

- Reflection
 - Rotation
 - Translation
 - Enlargement
-

Problem-Solving Strategies**RUCSAC Method:**

- **R**ead the problem carefully
- **U**nderline key information
- **C**hoose the operation/method
- **S**olve the problem
- **A**nswer the question asked
- **C**heck your answer makes sense

Estimation:

- Round numbers to check reasonableness
 - Use approximations for quick checks
 - Sense-check final answers
-

Common Real-World Contexts**Financial:**

- Bank statements and transactions
- Budgeting and expenses
- Interest calculations
- Discounts and sales
- GST calculations

Travel and Navigation:

- Distance, speed, time
- Fuel consumption

- Reading timetables
- Map interpretation

Cooking and Recipes:

- Scaling quantities
- Measurement conversions
- Ratio adjustments

Home and Work:

- Measuring for projects
 - Calculating materials needed
 - Understanding bills and invoices
-

Assessment Tips

1. **Read carefully:** Understand what the question asks
 2. **Show your working:** Even if answer is wrong, method may be correct
 3. **Check units:** Make sure your answer uses correct units
 4. **Estimate first:** Get a rough idea of what answer should be
 5. **Check reasonableness:** Does your answer make sense in context?
 6. **Use a calculator wisely:** Enter numbers correctly, don't rely solely on it
-

Common Student Mistakes

Arithmetic:

- Incorrect order of operations
- Decimal point errors
- Forgetting to convert units
- Calculator entry errors

Problem Solving:

- Not reading the question fully
- Choosing wrong operation
- Not answering what was asked
- Forgetting to include units

Statistics:

- Misreading graphs
 - Confusing mean, median, mode
 - Drawing unsupported conclusions
-

NZ Curriculum Alignment

The Numeracy co-requisite aligns with approximately Level 4/5 of the NZ Curriculum (Phase 3/4).

Students should:

- Have full control over Phase 3 numeracy skills
 - Be beginning to work at Phase 4
-

Assessment Information

CAA Format:

- Digital-first (completed on computer)
- Approximately 60 minutes (no strict time limit in single session)
- Multiple attempts allowed (no maximum resits)
- Available from Year 9 onwards

Allowed Materials:

- Standard calculator (non-scientific, non-graphing)
- Ruler
- Blank paper for workings (provided by school)
- Headphones for text-to-speech (Polly)

Not Allowed:

- Phones
 - Notes
 - Scientific/graphing calculators
 - Communication with others
-

References

- NZQA Numeracy Standards: nzqa.govt.nz/ncea/subjects/litnum/
- US32406 Details: nzqa.govt.nz/ncea/assessment/view-detailed.do?standardNumber=32406
- NCEA Education: ncea.education.govt.nz/literacy-and-numeracy/numeracy
- Past Papers: nzqa.govt.nz/ncea/subjects/past-exams-and-exemplars/litnum/32406/